

15-663 Computational Photography

Homework 2: HDR Imaging

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1 HDR imaging

Develop RAW images

I used the following command to convert RAW images to linear 16-bit TIFFs:

```
dcraw -w -q 3 -o 1 -4 -T image.NEF
```

The flags select camera white balance (`-w`), high-quality AHD demosaicing (`-q 3`), sRGB output (`-o 1`), linear 16-bit output (`-4`), and TIFF format (`-T`).

Linearize rendered images

Figure 1 shows the recovered log inverse response function g . JPEG intensities are linearized using $I_{\text{lin}} = \exp(g(I))$ before merging.

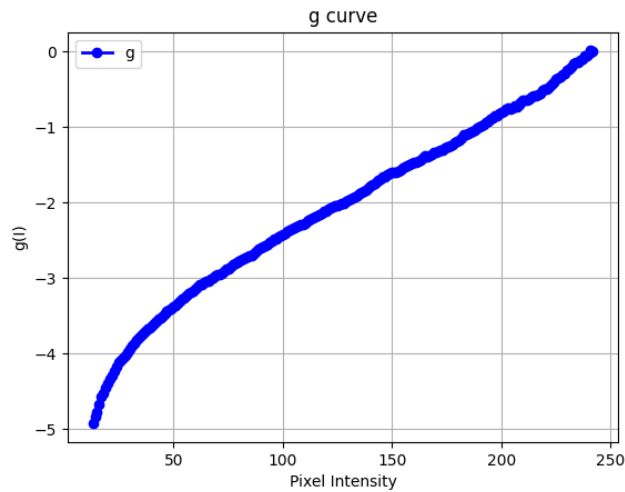


Figure 1: Recovered log inverse response function for the rendered images.

Merge the exposure stack

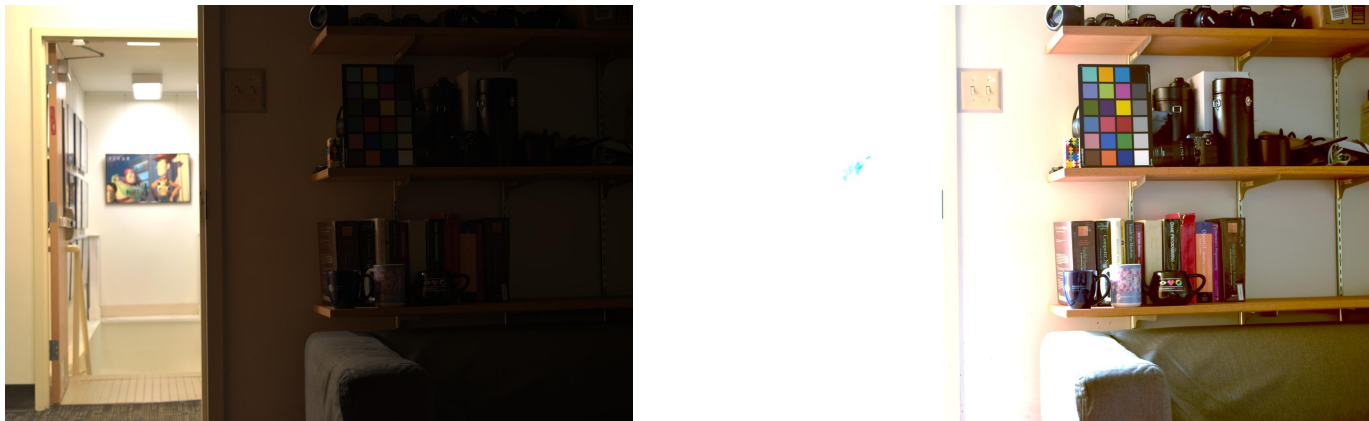
I used $Z_{\min} = 0.05$ and $Z_{\max} = 0.95$. My preferred result uses the JPEG stack, logarithmic merging, and Gaussian weighting (Figure 2a). The differences among the merging methods and weighting functions were visually subtle, so this selection is based mainly on overall appearance.

In the TIFF-based results, I observed yellow artifacts in the shadows. A possible cause is that blue-channel values fall below $Z_{\min}$ and receive zero weight; this explanation would need to be confirmed by inspecting the validity masks and the handling of zero total weight.

2 Color correction and white balancing

I fitted an affine transform to 24 color-checker patch averages and their reference RGB values, clipped negative results to zero, and scaled red and blue relative to green using a selected white patch.

I prefer the warmer original (Figure 2). The corrected preview looks cooler and has a clipped doorway. Matching the preview brightness would make the color comparison more reliable.



(a) Selected JPEG HDR preview.

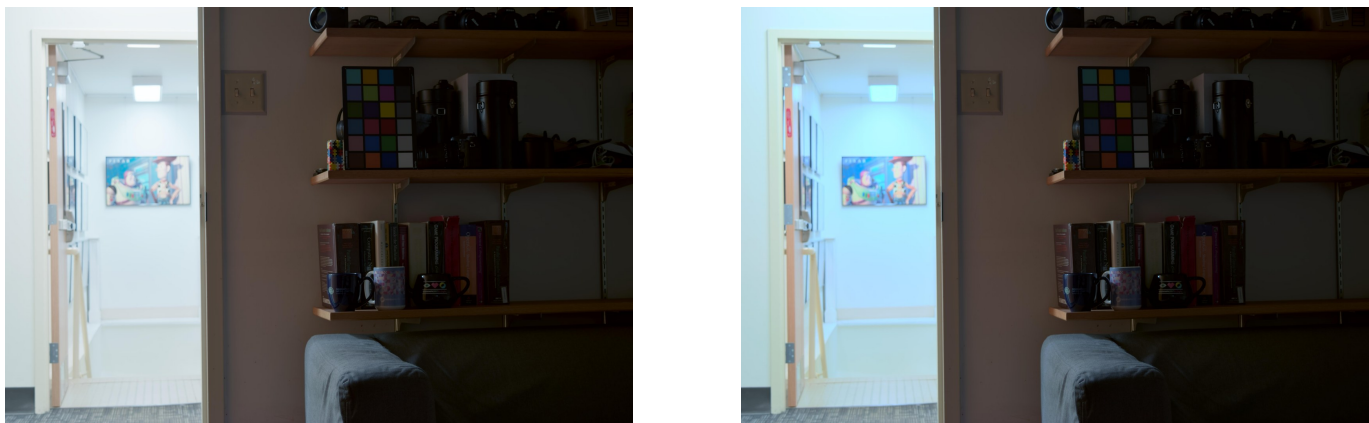
(b) Color-corrected preview.

Figure 2: Original and corrected previews; visible clipping may arise from display scaling.

3 Photographic tonemapping

Figure 3 compares photographic tonemapping applied jointly to RGB with tonemapping applied only to luminance Y in xyY . I prefer the RGB result because the bright light appears more neutral; the luminance-only result has a stronger blue appearance in that region.

The settings recorded in my original report are $K = 0.05$ and $B = 0.95$.



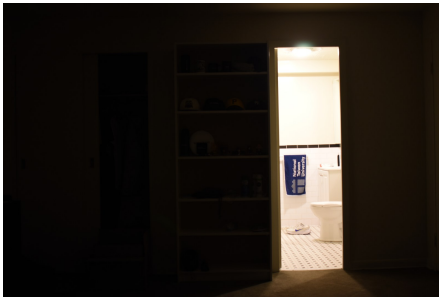
(a) Tonemapping jointly in RGB.

(b) Tonemapping luminance in xyY .

Figure 3: Comparison of the two photographic tonemapping methods.

4 Create and tonemap my own HDR photo

I photographed a bright bathroom beside a dark room (Figure 4), then merged the RAW/TIFF and linearized JPEG stacks separately using logarithmic merging and Gaussian weights.



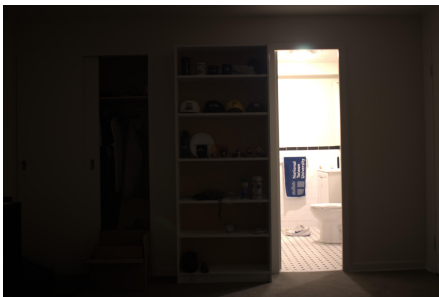
(a) Input exposure 10.



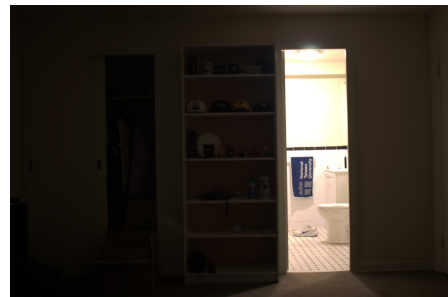
(b) Input exposure 12.

Figure 4: Two JPEG exposures from my captured stack.

Figure 5 explicitly compares the two source formats. Both previews retain bathroom detail while the adjacent room remains dark. Their display scaling differs, so this is not a controlled brightness comparison.



(a) RAW/TIFF-derived HDR preview.



(b) JPEG-derived HDR preview.

Figure 5: Logarithmic merging with Gaussian weights. These are scaled, gamma-encoded previews, not photographic tonemaps.

The photographic tonemaps in Figure 6 reveal more detail in the clothing and shelving. I prefer the RGB rendition visually, although the luminance-only result better matches my recollection of the yellow illumination.



(a) Photographic tonemapping in RGB.



(b) Photographic tonemapping in xyY .

Figure 6: Photographic tonemapping comparison for my scene.

5 Noise calibration and optimal weights

Noise calibration

I randomly selected four G1 pixels and plotted their values across 50 ramp frames after subtracting the average dark frame (Figure 7). The histograms approximately follow a Gaussian distribution, consistent with the Gaussian approximation to the camera noise model at sufficient signal levels.

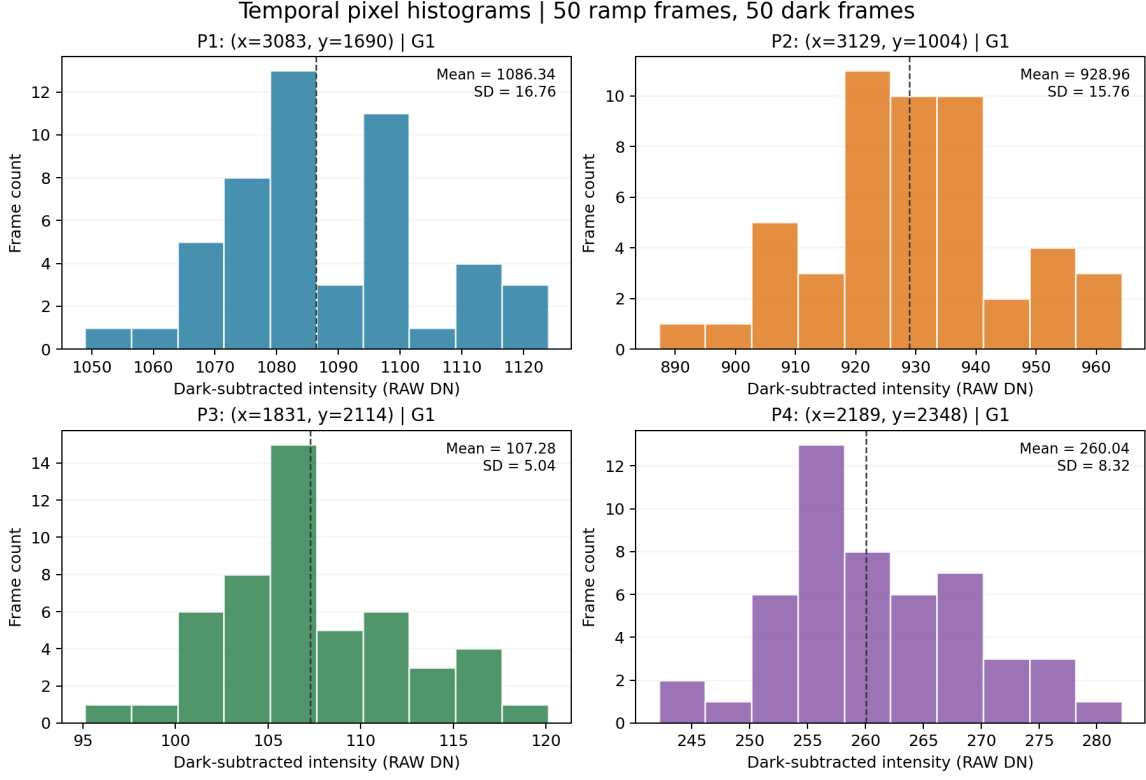


Figure 7: Temporal intensity histograms for four G1 pixels. Dashed lines indicate sample means; SD uses the $N - 1$ denominator. Each histogram contains 50 observations.

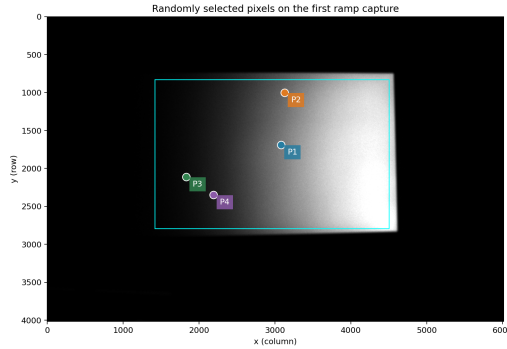


Figure 8: Selected pixels on the first ramp frame; the cyan rectangle marks the sampling crop. Coordinates are zero-based full-RAW image coordinates.

Across repeated trials, the fitted slope g was relatively stable, whereas the intercept varied. Figure 9 shows an example fit for the G1 channel at ISO 200. The estimated gain is $g = 4.146$ and the fitted additive-noise variance is 56.47 in scaled TIFF DN/electron and scaled TIFF DN², respectively. Table 1 reports the results across ISO settings; the fitted intercept and measured dark-frame variance are distinct estimates.

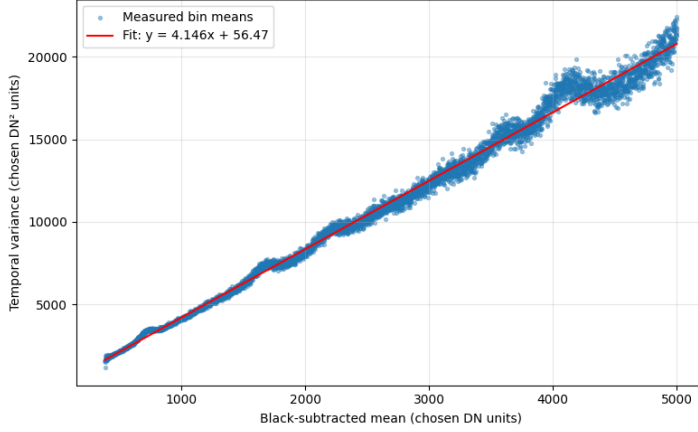


Figure 9: Mean-variance fit for G1 at ISO 200.

ISO	g	Intercept	Dark var.
200	4.146	56.47	156.1
400	8.268	402.0	409.0
800	16.58	1189	1275
1600	32.56	5520	4364
3200	65.14	21096	16571

Table 1: G1 noise-calibration results.

Merging with optimal weights

Figure 10 shows the preview from the optimal-weight RGB merge. I used a shared white balance across exposures to keep the weighting inputs consistent. At this display size, I do not see a large improvement. A comparison of matching shadow crops under identical tonemapping settings is needed to assess noise reduction reliably.



Figure 10: Tonemapped preview of the optimal-weight RGB merge.

6 Bonus: HDR by varying both shutter speed and ISO

Capture setup

I captured the same room at $f/4$ using the two schedules in Table 2. The fixed-ISO bracket uses ISO 200 throughout, while the second bracket increases ISO for the longer exposures. I reused the first three ISO 200 frames in both brackets. Each effective bracket contains eight frames with a total exposure time of 10.6688 seconds. This equalizes shutter-open time; the manually captured brackets do not have equal measured elapsed capture time. I selected the schedule directly rather than implementing the capture optimization in Section 4.2 of Hasinoff et al.

Frame	1	2	3	4	5	6	7	8
Shutter (s)	1/2000	1/500	1/125	1/30	1/8	1/2	2	8
Fixed ISO	200	200	200	200	200	200	200	200
Varying ISO	200	200	200	400	800	800	1600	1600

Table 2: Capture schedules with identical shutter times and frame counts.

Calibration and ISO-aware merging

For each ISO and Bayer channel (R, G1, G2, B), I estimated the gain g_k from the ramp mean-variance slope and measured the additive variance $\text{var}_{\text{dark},k}$ directly from repeated black frames. I used the measured black variance, not the intercept of the ramp fit. The parameters are saved in `iso_noise_profile.json` for reuse. Table 3 uses the same scaled TIFF units as Table 1: $S = 65535/(4095 - 150)$, $g_{\text{scaled}} = Sg_{\text{RAW}}$, and $\text{var}_{\text{scaled}} = S^2\text{var}_{\text{RAW}}$. Small differences remain because the bonus calibration uses a different sampling stride.

ISO	200	400	800	1600
g (scaled DN/electron)	4.136	8.267	16.594	32.597
var_{dark} (scaled DN^2)	154.1	405.8	1255.6	4325.0

Table 3: Measured G1 parameters expressed in scaled TIFF units (scaling only, before white balance).

Following the inverse-variance approach in Hasinoff et al., Section 4.1, let z_k be the dark-subtracted signal in scaled TIFF DN, with g_k and $\text{var}_{\text{dark},k}$ in matching scaled units. Using gain in DN/electron (the reciprocal of the paper’s convention), I compute

$$L_k = \frac{z_k}{g_k t_k}, \quad \text{var}_k = g_k \max(z_k, 0) + \text{var}_{\text{dark},k}, \quad (1)$$

$$w_k = \frac{(g_k t_k)^2}{\text{var}_k}, \quad L_{\text{HDR}} = \frac{\sum_k w_k L_k}{\sum_k w_k}. \quad (2)$$

The implementation computes the equivalent expressions in native RAW units. Samples at 0 or at/above 4000 RAW DN receive zero weight. Signed dark-subtracted values are retained when averaging; the maximum with zero is used only for predicting variance. I merge in patches before demosaicing, using the same method for both brackets.

I use empirical per-ISO additive variances instead of separately estimating σ_{read}^2 and σ_{ADC}^2 , so that additional bonus calibration requirement remains incomplete. Dark correction uses channel-average offsets from short-exposure black frames; exposure-dependent dark current is not modeled.

Results and comparison

Figure 11 compares the merged images. Both use the same white balance and display scalar, followed by clipping and sRGB gamma encoding. Their linear means before clipping are 0.2250 and 0.2234. The previews use bilinear demosaicing without a camera-to-sRGB color matrix.



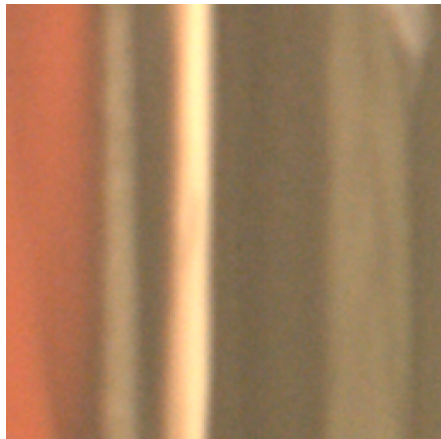
(a) Fixed ISO 200.



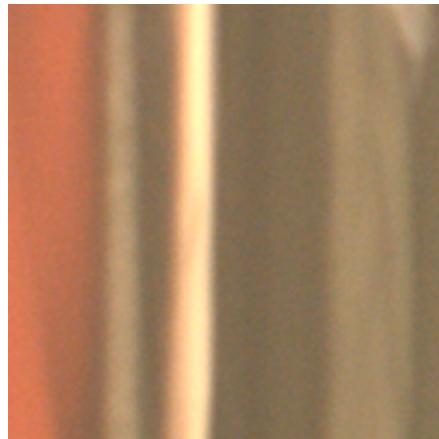
(b) Varying ISO 200–1600.

Figure 11: HDR previews from the two equal-integration-time brackets.

I cannot distinguish a clear noise improvement at normal viewing size. Even after brightening and enlarging the same dark clothing crop (Figure 12), both results show similar grain. The gain/noise model predicts only about 0.5% lower noise standard deviation on a sampled dark wall region and 2.2% on the clothing region. These are model predictions, not measured reductions from repeated brackets. The long exposures collect enough light that shot noise dominates these regions, limiting the benefit of higher ISO.



(a) Fixed ISO.



(b) Varying ISO.

Figure 12: The same 256×256 -pixel clothing region, enlarged with nearest-neighbor sampling. A shared local display scalar brightens both crops equally.

The varying-ISO crop also has a small brightness offset, and the two brackets share three input frames. Therefore, their difference image alone cannot establish a noise reduction. I do not have a clear visual preference between these two results; repeated independent captures would be needed to measure such a small advantage reliably.